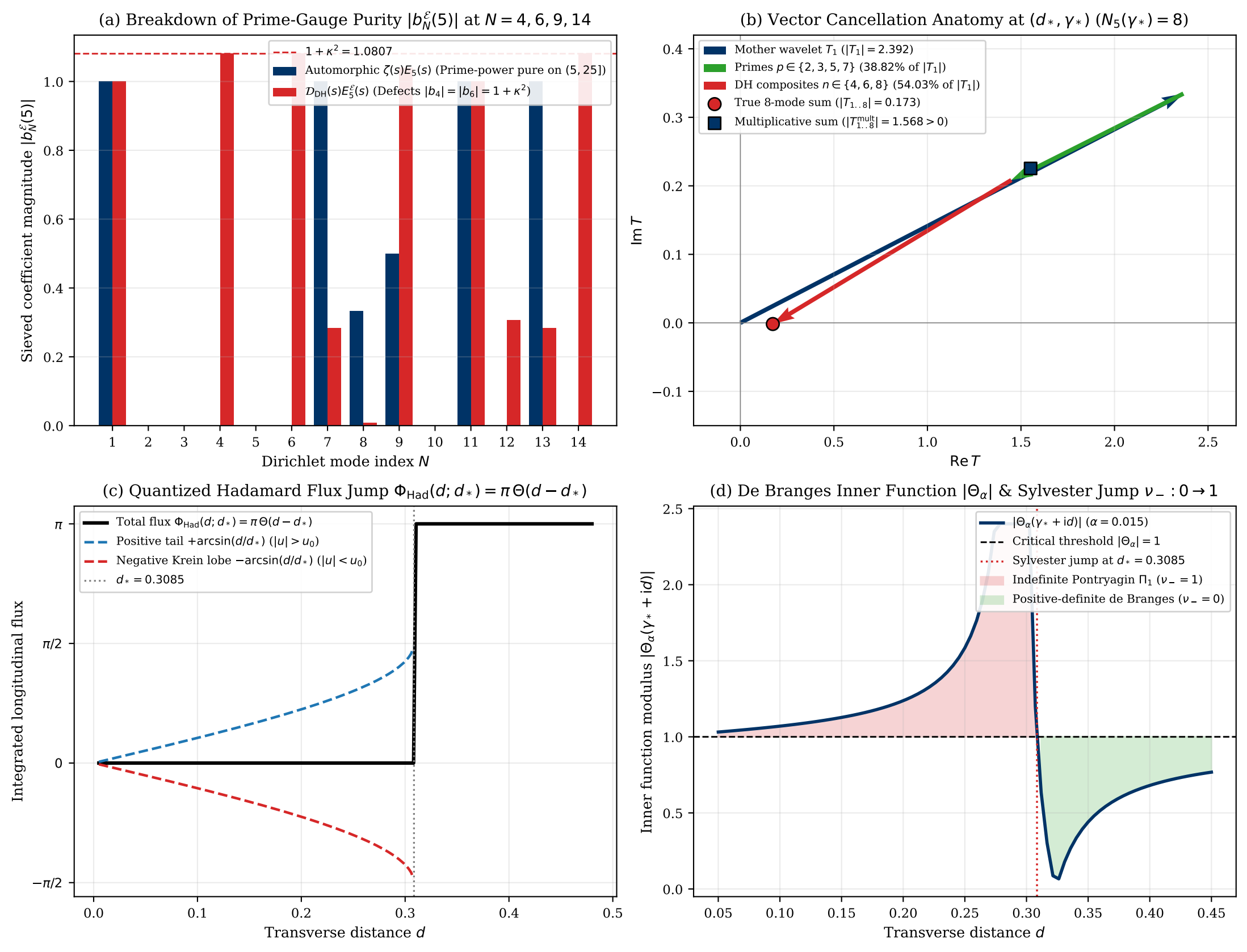

(a) Breakdown of Prime-Gauge Purity $|b_N^{\mathcal{E}}(5)|$ at $N = 4, 6, 9, 14$
$1 + \kappa^2 = 1.0807$
Automorphic $\zeta(s)E_5(s)$ (Prime-power pure on $(5, 25]$)
$\mathcal{D}_{\mathrm{DH}}(s)E_5^{\mathcal{E}}(s)$ (Defects $|b_4| = |b_6| = 1 + \kappa^2$)
Sieved coefficient magnitude $|b_N^{\mathcal{E}}(5)|$
Dirichlet mode index N
(b) Vector Cancellation Anatomy at (d_*, γ_*) $(N_5(\gamma_*) = 8)$
Mother wavelet T_1 ($|T_1| = 2.392$)
Primes $p \in \{2, 3, 5, 7\}$ (38.82% of $|T_1|$)
DH composites $n \in \{4, 6, 8\}$ (54.03% of $|T_1|$)
True 8-mode sum ($|T_{1..8}| = 0.173$)
Multiplicative sum ($|T_{1..8}^{\mathrm{mult}}| = 1.568 > 0$)
$\mathrm{Im}\, T$
$\mathrm{Re}\, T$
(c) Quantized Hadamard Flux Jump $\Phi_{\mathrm{Had}}(d; d_*) = \pi\Theta(d - d_*)$
Total flux $\Phi_{\mathrm{Had}}(d; d_*) = \pi\Theta(d - d_*)$
Positive tail $+\arcsin(d/d_*)$ ($|u| > u_0$)
Negative Krein lobe $-\arcsin(d/d_*)$ ($|u| < u_0$)
$d_* = 0.3085$
Integrated longitudinal flux
Transverse distance d
(d) De Branges Inner Function $|\Theta_\alpha|$ & Sylvester Jump $\nu_- : 0 \to 1$
$|\Theta_\alpha(\gamma_* + id)|$ ($\alpha = 0.015$)
Critical threshold $|\Theta_\alpha| = 1$
Sylvester jump at $d_* = 0.3085$
Indefinite Pontryagin Π_1 ($\nu_- = 1$)
Positive-definite de Branges ($\nu_- = 0$)
Inner function modulus $|\Theta_\alpha(\gamma_* + id)|$
Transverse distance d